

AIDSR: Agentic & AI-assisted Data Science with R

Full Program, September-November 2026 Cohort

Imad EL BADISY, PhD

1 Course Overview

Title: Agentic & AI-assisted Data Science with R

Format: Online, cohort-based

Duration: 2 weeks, 6 sessions, 2 hours per session, 12 contact hours

Dates: 12/10/2026 – 23/10/2026

Days: Monday, Wednesday, Friday (18h00 to 20h00)

Time: Morocco time

Price: 150€ (1500 MAD)

Instructor: Imad EL BADISY, PhD

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1.1 Course Description

AIDSR extends the MHDSR pathway into agentic and AI-assisted data science with R: using a command-line coding agent to drive a reproducible analysis and ML pipeline under version control, and using large language models for structured extraction and text classification in health research.

The course is hands-on and opinionated about guardrails: version control, deterministic and logged runs, evaluation against a gold standard, cost and reproducibility, privacy, and a clear sense of when not to use an LLM or an agent.

1.2 Target Audience

Researchers, PhD students, clinicians, public health professionals, and data analysts.

1.3 Prerequisites

ML4HOR (or equivalent machine-learning experience) and R at the level covered in BIOSTATR.

1.4 Learning Objectives

By the end of AIDSRS, participants will be able to:

1. set up and drive a command-line coding agent on a version-controlled analysis project;
2. use an agent to generate, execute, and review analysis code reproducibly;
3. build an end-to-end, reproducible ML/analysis pipeline with agent assistance;
4. design schemas and prompts for structured extraction from clinical text and documents;
5. build and evaluate LLM-based text classification and annotation workflows;
6. assess LLM outputs against a gold standard, and quantify agreement, cost, and error;
7. apply guardrails for determinism, logging, privacy, and reproducibility;
8. judge when an LLM or agent is, and is not, appropriate for a task.

2 Detailed Schedule

Table 1: AIDSRS detailed schedule

No.	Week	Date	Time	Topic	Main Content
1	1	12/10/2026 (Mon)	18h00- 20h00	Agentic workflows and CLI coding agents	What a coding agent is, setup, project structure, version control, permissions and guardrails, safe iteration
2	1	14/10/2026 (Wed)	18h00- 20h00	Code generation, execution and review with an agent	Iterating on analysis code, tests, reproducible environments, reviewing and correcting agent output
3	1	16/10/2026 (Fri)	18h00- 20h00	A reproducible pipeline end to end	From raw data to report with agent assistance, under version control, with logging and re-runs
4	2	19/10/2026 (Mon)	18h00- 20h00	LLMs for structured extraction	Schema design, prompting, few-shot examples, validation, evaluation against a gold standard
5	2	21/10/2026 (Wed)	18h00- 20h00	LLMs for text classification and annotation	Zero/few-shot classification, prompt design, inter-rater agreement, bias and failure modes
6	2	23/10/2026 (Fri)	18h00- 20h00	Guardrails, evaluation and reproducibility	Determinism, logging, cost, privacy, error analysis, when not to use an LLM, final mini-project

3 Course Materials

- Slides and self-contained notebooks distributed before each session
- R scripts, datasets, prompt templates, and exercises (shared repository)
- Exercises reviewed and corrected during sessions

4 Assessment

The course ends with a short reproducible Quarto report or mini-project, presented and discussed in the final session.

5 About the Instructor

Imad EL BADISY, PhD, is a biostatistician working on statistical and machine learning methods for health data, with a focus on survival analysis. Full profile, publications, and R packages: elbadisyimad.com

6 Policies

Language: Teaching in French, with English for statistical and methodological terminology.

Software: R (≥ 4.4) and RStudio required; a command-line coding agent and access to an LLM API are used in class.

Attendance: Full attendance at all 6 sessions is recommended given the cumulative structure.